Jitter fitting and BER extrapolation

HW13 - Signal integrity numerical analysis

The verified full-length HW12 data give a two-Gaussian tail-model minimum BER of **2.94553e-13 for Case 3 Rx** and **2.30408e-08 for Case 4 Rx**. At BER = 1e-12, the nominal Case 3 eye width is **0.830023 ps (0.0332009 UI)**; Case 4 has no opening. These are extrapolated model predictions, not measured error rates. The sensitivity analysis on the final page shows why the narrow Case 3 opening must be interpreted cautiously.

Scope and provenance

All eight sampled CDFs and bathtubs are calculated for four cases at Tx and Rx. The two-Gaussian extrapolation then focuses on Case 3 and Case 4 Rx, as requested in the assignment [1]. The primary input is the saved, independently verified HW12 horizontal eye-slice distribution from **1,048,576 evaluated bits per case**. That full channel simulation was reused; **HW12 was not rerun for this analysis**.

Physical / numerical quantity	Setting
Signaling / UI	40 Gb/s NRZ / 25 ps
Cases 1-4	No Tx jitter; SJ; RJ; SJ + RJ
SJ / RJ	3 ps peak at 1 MHz / zero-mean Gaussian, sigma = 1 ps
Source / load	20 ohm / 85 ohm, each with 0.3 pF shunt capacitance
Rise and fall duration	5 ps
Statistical time grid	200 bins per UI; dt = 0.125 ps
Transition density	rhoT = 0.5
Adjacent ideal boundaries	t_i = 12.5 ps; t_(i+1) = 37.5 ps

Probability definitions and timing convention

For normalized bin probability p[k], the PMF is dimensionless and sums to one. The density f[k] = p[k]/dt has units 1/ps and integrates to one. The empirical CDF at a bin center uses half that cell's mass; interpolation assumes uniform density inside each centered 0.125 ps cell. This fixes the time origin and keeps the CDF and density consistent, rather than treating endpoint-inclusive samples as a different physical grid.

```
F(u) = P(jitter <= u);  S(u) = P(jitter > u)
BER(t) = rhoT * [ S(t - 12.5 ps) + F(t - 37.5 ps) ]
f2G(u) = tau1 * phi((u-mu1)/sigma1)/sigma1
       + tau2 * phi((u-mu2)/sigma2)/sigma2
F2G(u) = tau1 * Phi((u-mu1)/sigma1)
       + tau2 * Phi((u-mu2)/sigma2);  tau2 = 1 - tau1
```

Here phi and Phi denote the standard-normal PDF and CDF. The timing-error coordinate u is centered at an ideal transition; absolute sampling time t lies between the two boundaries. Gaussian SF and CDF tails are evaluated directly with stable log-domain normal-CDF/SF functions and log-sum addition, including the SF instead of subtracting a nearly unit CDF from one [2].

Reproduce from the repository root

```
python homework/hw13/verify_inputs.py
python homework/hw13/src/hw13.py
python -m unittest discover -s homework/hw13/tests -v
python homework/hw13/tests/verify_results.py
python -m pip install -r homework/hw13/report/requirements.txt
python homework/hw13/report/build_report.py --refresh
```

1. Sampled distributions: all eight cases

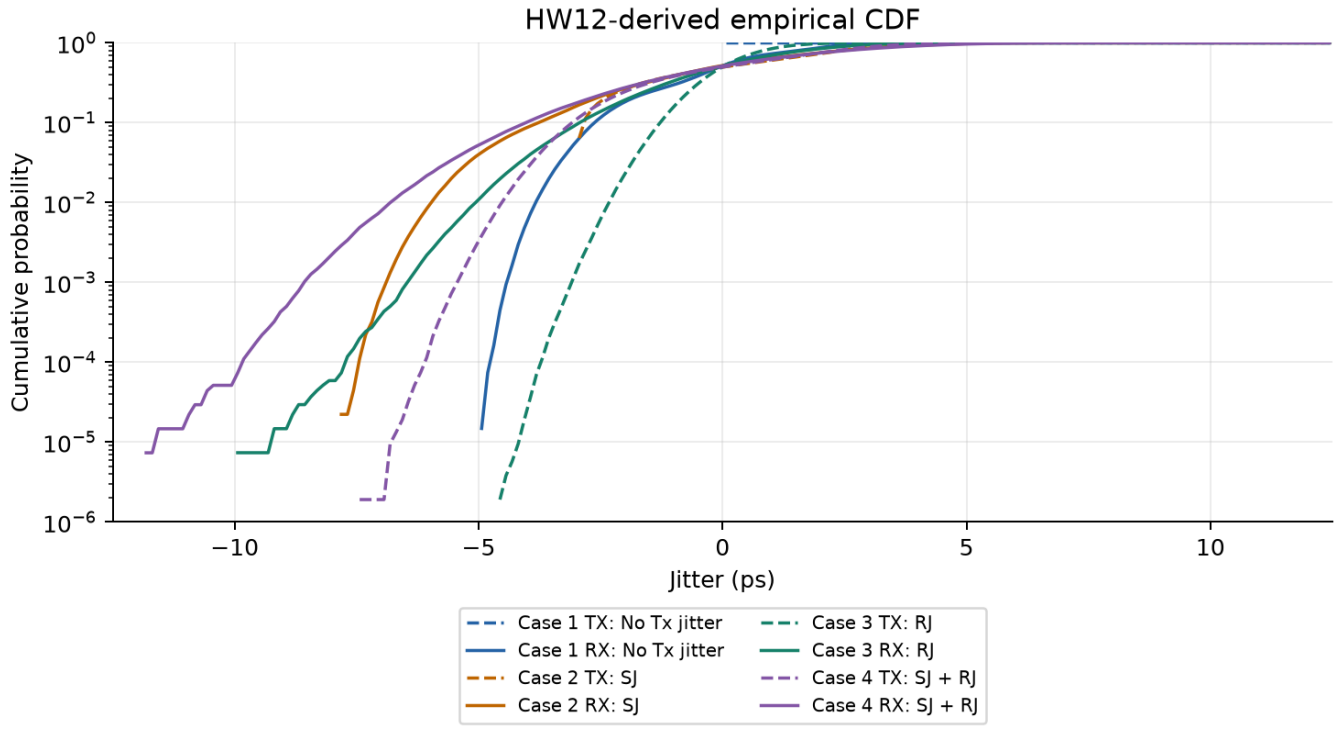


Figure 1. Sampled CDFs for every Tx/Rx distribution. Color identifies the case; dashed and solid lines distinguish Tx and Rx. Zero CDF values are masked only for logarithmic display. No low-probability floor is added to the analysis.

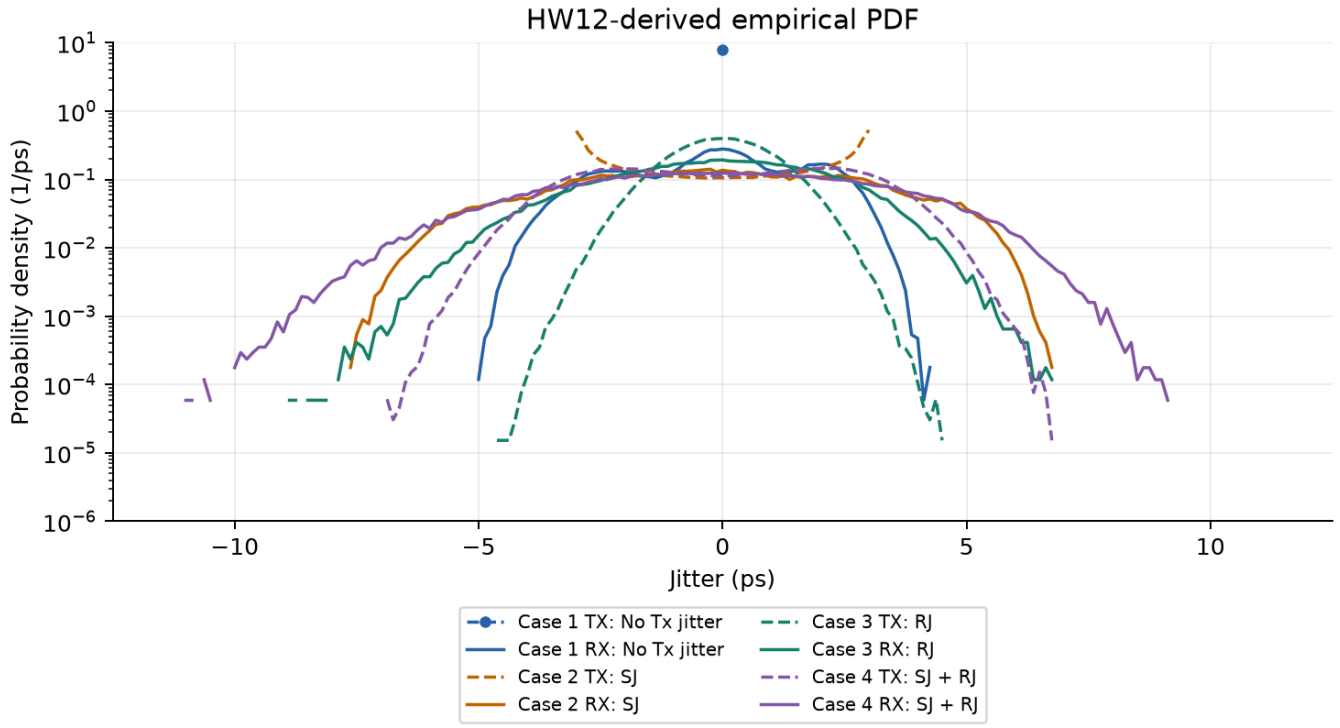


Figure 2. The same eight distributions as PDFs in 1/ps. These are normalized eye-slice densities, not counts or probabilities per bin. The narrow ideal-Tx peak is a sampled, quantized distribution.

The CDF is an accumulated PMF, not an accumulated PDF without a dt factor. Case 1 Rx spreads despite its ideal Tx because channel ISI causes DDJ. Adding SJ widens the distributions; RJ creates sampled tails. The central multimodal Rx structure need not have a Gaussian shape and is excluded from the tail fit. The eye slice also weights voltage occupancy and edge slope; it is not an equal-weight histogram of independently detected crossing events [2, pp. 2-4].

2. Sampled bathtub curves and finite evidence

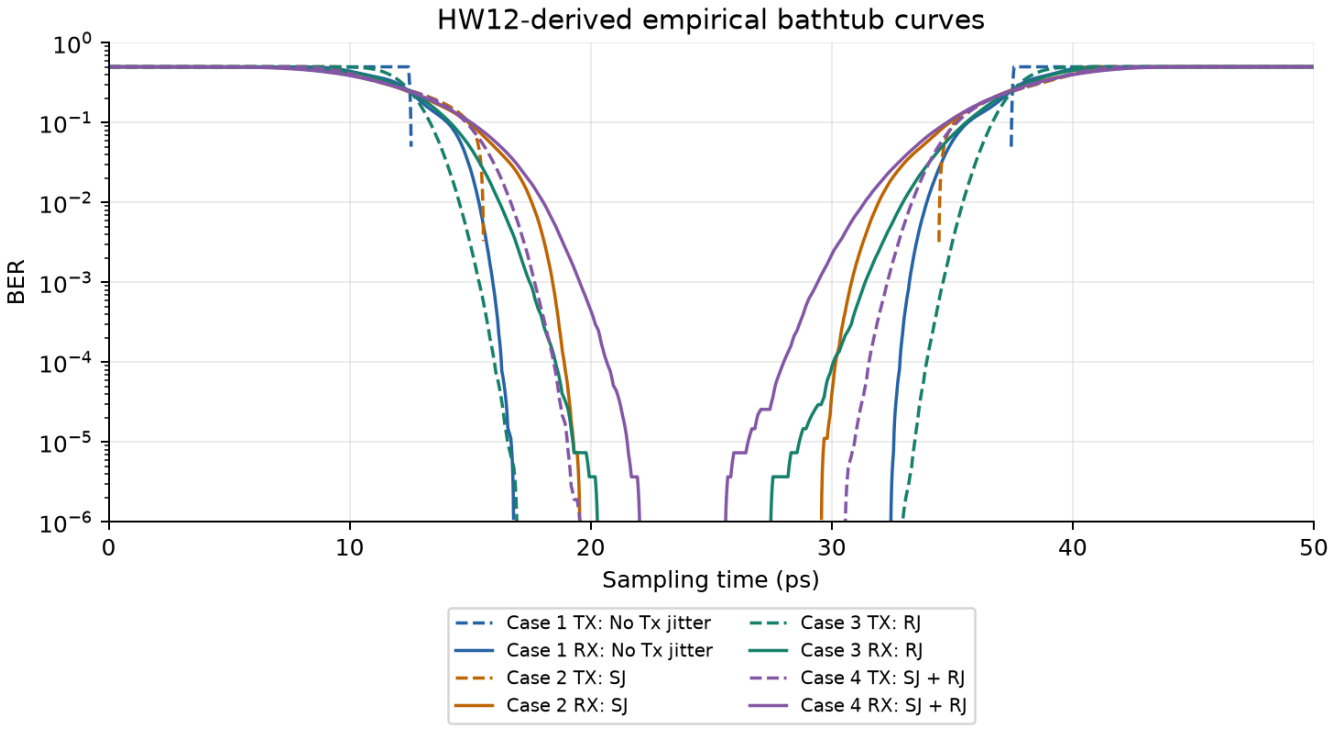


Figure 3. All eight non-extrapolated bathtub curves, formed from the empirical CDF and survival function at neighboring transitions. Curves cover the 2-UI plotting interval; eye-width metrics use the eye between 12.5 and 37.5 ps.

The late left transition contributes S(t - t_i); the early right transition contributes F(t - t_(i+1)). Moving the sampler toward either boundary increases its corresponding tail. This is the two-transition construction in the lecture [2, pp. 10-13]. The left PDF tail governs the right side of the bathtub, and the right PDF tail governs its left side [1, p. 2].

Distribution	Slice observations	Zero bins	Smallest nonzero PMF
Case 1 TX	524,108	199	1
Case 1 RX	134,840	125	7.4162e-06
Case 2 TX	524,108	151	0.0130469
Case 2 RX	134,742	82	1.48432e-05
Case 3 TX	524,108	126	1.908e-06
Case 3 RX	135,727	72	7.36773e-06
Case 4 TX	524,108	87	1.908e-06
Case 4 RX	136,081	39	7.34856e-06

All eight PMFs sum to one and all eight PDFs integrate to one on the recorded grid. The receiver slice counts differ from the bit count: Case 3 and Case 4 contain 135,727 and 136,081 threshold-voltage occupancies. Their smallest nonzero PMF increments are approximately 7.4e-6. Thus a zero empirical bathtub value is absence of observed slice support, **not evidence of zero physical BER or direct observation of BER = 1e-12**. The zeros remain exact in numerical arrays and are omitted only when plotting on logarithmic axes.

3. Tail model and Case 3 Rx result

The two-Gaussian model has separate means, standard deviations and a single independent mixing weight. Only tail locations with one-sided empirical CDF or SF in **[3e-4, 3e-2]** are selected. Joint nonlinear least squares minimizes unweighted residuals in log10(CDF) and log10(SF); it does not fit the central DJ peak. Six starting guesses converge, and neither reported solution reaches a parameter bound. This is a tail-fit procedure, **not maximum likelihood**.

The fit evaluates a selected bin's CDF at its upper edge, u[k] + dt/2, and its SF at the lower edge, u[k] - dt/2, so each includes the full selected cell mass. This half-bin correction is distinct from the center-interpolated plotted CDF. The bounds are mu1 in [-8, 0] ps, mu2 in [0, 8] ps, both sigmas in [0.2, 5] ps, and tau1 in [0.01, 0.99].

Tail-model parameter	Case 3 Rx	Case 4 Rx
mu1 (ps)	-1.94537	-3.97865
sigma1 (ps)	1.6379	1.8081
mu2 (ps)	0.939036	2.00569
sigma2 (ps)	1.47729	1.78834
tau1	0.35408	0.171466
tau2 = 1 - tau1	0.64592	0.828534

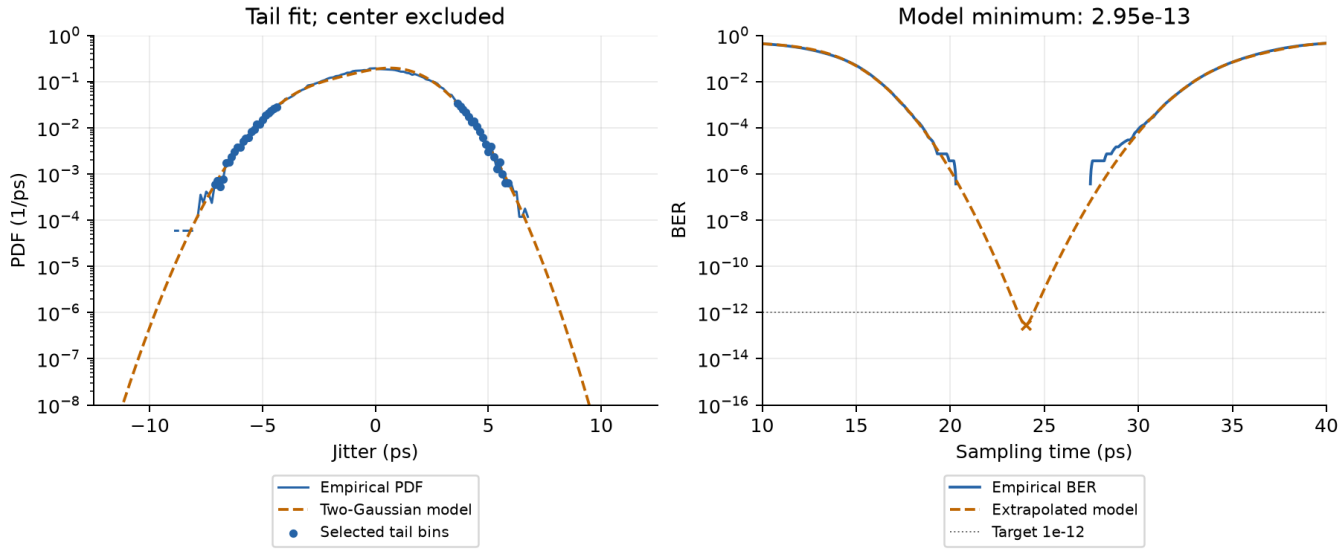


Figure 4. Case 3 Rx: sampled PDF versus the two-Gaussian model and sampled bathtub versus extrapolation. Highlighted PDF points identify the selected tail bins; the model extends below the empirical evidence scale.

Fit diagnostic	Case 3 Rx	Case 4 Rx
Selected tail locations	42	51
CDF/SF log10 RMSE	0.0170397	0.011733
PDF-tail log10 RMSE	0.0703944	0.0605754
Converged starts	6 / 6	6 / 6

Case 3 fits left-center coordinates from -7.125 to -4.375 ps and right-center coordinates from 3.625 to 5.875 ps. The selected probability threshold corresponds to approximately 40.7 cumulative slice observations at its low end. The nominal minimum is **2.94553e-13** at **24.0038 ps**. At 1e-12, the crossing times are 23.6012 and 24.4312 ps, giving **0.830023 ps** of opening. The lower-than-empirical values follow the Gaussian tail assumption, not new simulated errors.

4. Case 4 Rx and low-BER eye widths

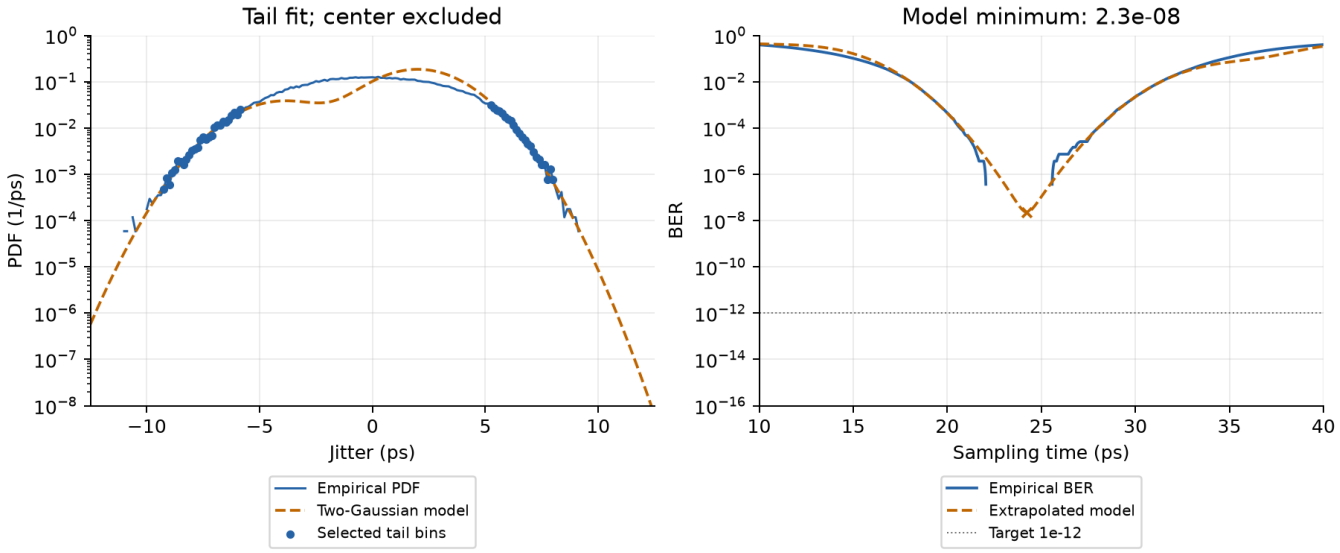


Figure 5. Case 4 Rx PDF-tail fit and extrapolated bathtub. The predicted minimum remains above 1e-9, so there is no modeled eye opening at 1e-9, 1e-12 or 1e-15.

Case 4 fits left-center coordinates from -9.25 to -5.875 ps and right-center coordinates from 5.25 to 8 ps. The nominal minimum is **2.30408e-08** at **24.231 ps**. An eye width exists only when the target BER is at least the curve minimum; it is the distance between the two target-BER crossings inside this eye.

Target BER	Case 3 EW (ps)	Case 3 EW (UI)	Case 4 EW (ps)
1e-06	8.24695	0.329878	3.19011
1e-09	4.17788	0.167115	No opening
1e-12	0.830023	0.0332009	No opening
1e-15	No opening	No opening	No opening

Corrections and comparison with supplied code

The executable analysis preserves zero probabilities, keeps dt in ps throughout, uses the physical 0.125 ps grid, and computes extreme Gaussian SF/CDF tails directly. It resolves data files relative to the project, uses valid positive logarithmic display limits, and calculates numerical minima and crossing roots. The incomplete starter is retained as instructions. Both supplied legacy variants contain an inherited 1e-15 display offset and inconsistent endpoint coordinates; those artifacts are not accepted as fresh tail observations.

Comparison / Case 3	Minimum BER	EW at 1e-12 (ps)
Verified HW12 input, tail fit	2.94553e-13	0.830023
Corrected legacy CSV, tail fit	3.49805e-13	0.762122
Archive variant, supplied parameters	2.06705e-12	No opening
Adjacent variant, supplied parameters	6.26121e-14	1.56811

The legacy comparison subtracts the known display offset and restores the physical grid before refitting. Its original observation counts and exact generating run are unavailable. The two supplied parameter sets also predict different Case 3 conclusions. They are references, not a unique numerical answer and not the primary input used above.

5. Sensitivity and limitations

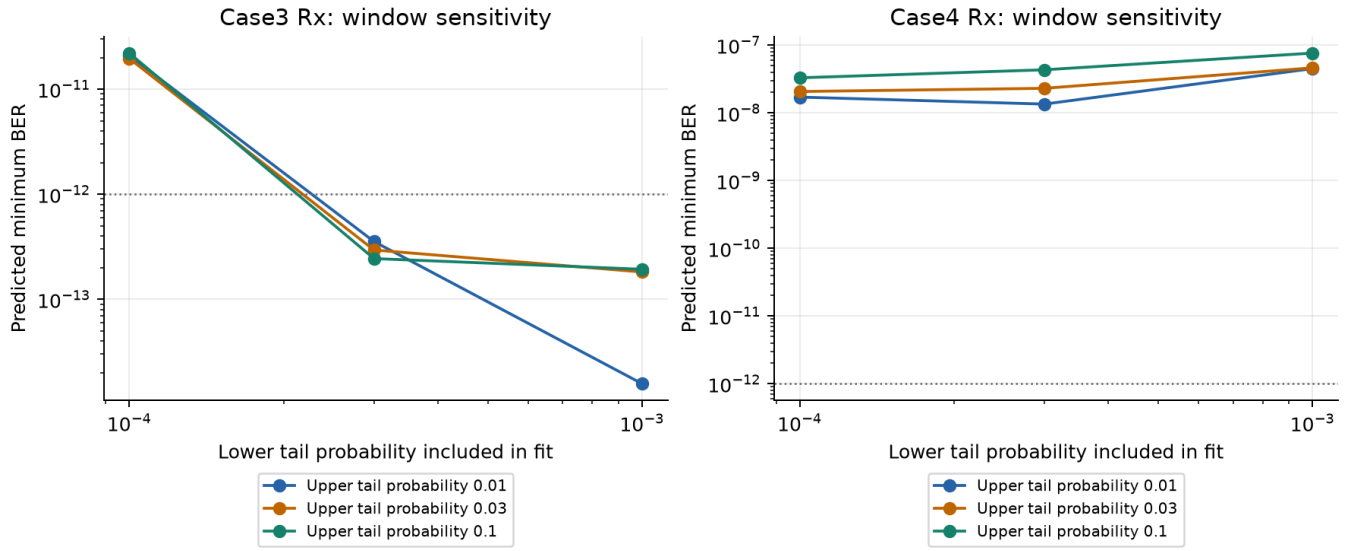


Figure 6. Predicted minimum BER for nine tail-window choices per receiver case. The table also gives Case 3 eye widths at 1e-12. These model responses do not represent confidence intervals.

Tail min	Tail max	Case 3 min BER	Case 3 EW (ps)	Case 4 min BER
0.0001	0.01	2.09283e-11	No opening	1.7093e-08
0.0001	0.03	1.9581e-11	No opening	2.06607e-08
0.0001	0.1	2.20289e-11	No opening	3.29366e-08
0.0003	0.01	3.54793e-13	0.749623	1.34737e-08
0.0003	0.03	2.94553e-13	0.830023	2.30408e-08
0.0003	0.1	2.44072e-13	0.908142	4.32417e-08
0.001	0.01	1.57253e-14	1.86983	4.50274e-08
0.001	0.03	1.81909e-13	1.0277	4.6204e-08
0.001	0.1	1.93038e-13	1.00421	7.62604e-08

Across all nine windows, Case 3 minimum BER ranges from 1.57253e-14 to 2.20289e-11, and its 1e-12 eye width ranges from no opening to 1.86983 ps. Including probabilities down to 1e-4 leaves only about 14 cumulative slice observations and can remove the opening; the narrow high-tail window can predict a larger opening. **These are fit-window sensitivity results, not confidence intervals.** The cumulative points are correlated and their RMSE is descriptive. Case 4 has no 1e-12 opening in all nine tested windows.

The Gaussian-tail and two-component assumptions, finite time/voltage bins, one-UI folding of extreme jitter, and slope-weighted eye slices limit physical accuracy. This report contains no low-BER measurement, hardware validation or new ADS ChannelSim result. Saved-artifact and numerical checks validate the stated calculation, not the model assumptions. All **15 tests passed**. An independent verifier, without importing the production module, checked 3,200 empirical bins, 26 model sets and 40,010 bathtub ordinates using direct math.erfc tails, cell-mass integration and separate minimum/root solvers. Maximum relative model-curve disagreement was 1.74943e-14.

References

[1] Homework: Jitter fitting and BER extrapolation, supplied assignment, pp. 1-2 (required tasks, finite-pattern note, zero-floor rule and tail mapping). [2] Jitter Basics (2), course handout 13, pp. 2-4 (eye slice and CDF), pp. 10-13 (two-boundary BER and eye width), pp. 15-17 (tail extrapolation and two-Gaussian mixture). The optional ADS example is not presented as a new result.